

Radar and Acoustic Sensor Fusion Using a Transformer Encoder for Robust Drone Detection and Classification

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Abstract—Unmanned aerial vehicles are increasingly deployed across diverse domains including security surveillance, precision agriculture, infrastructure inspection, and logistics. However, their rapid proliferation raises critical concerns regarding unauthorized intrusions, smuggling, and adversarial exploitation. Robust drone detection and classification systems are therefore essential. In this paper, we propose a multi-modal framework that fuses frequency-modulated continuous-wave radar and acoustic sensing using a transformer encoder-based architecture. Unlike single-modality approaches, our method exploits cross-modal complementarity to improve robustness against environmental noise, false alarms, and stealthy drone operations. Notably, our approach directly processes raw acoustic waveforms without domain transformations such as spectrograms, reducing preprocessing overhead and improving real-time applicability. Experimental results on our publicly released dataset demonstrate that the proposed method achieves excellent detection and classification accuracy under clean conditions, significantly outperforming existing state-of-the-art approaches while using fewer parameters.

Index Terms—UAV detection, UAV classification, sensor fusion, FMCW radar, acoustic sensing

I. INTRODUCTION

Unmanned aerial vehicles (UAVs), commonly known as drones, have undergone rapid advancements over the past decade, enabling applications across diverse sectors. In precision agriculture, drones facilitate crop health monitoring through multispectral imaging, precision pesticide spraying, irrigation management, and yield estimation [1], [2]. In the security domain, they support border patrol, crowd surveillance, search and rescue missions, and disaster response coordination. Infrastructure inspection including power lines, bridges, and wind turbines, cinematography, package delivery logistics, and environmental monitoring further illustrate the breadth of drone utility in modern society.

However, this rapid proliferation introduces significant security and privacy concerns. Unauthorized drone intrusions into restricted zones including airports, government facilities, military installations, nuclear power plants, and agricultural estates pose severe threats to public safety and national security [3], [4]. The consequences range from airspace disruption

and collision risks to espionage and physical damage. Drones have also been exploited for smuggling contraband into prisons, unauthorized surveillance, targeted attacks, and disruption of critical infrastructure. These threats span civilian, commercial, and military contexts, further underscoring the urgent need for reliable detection and mitigation strategies [5].

Motivated by these challenges, this work proposes a multi-modal drone detection and classification framework leveraging the complementary strengths of radar and acoustic sensing. Unlike single-modality approaches where acoustic systems suffer from environmental noise and radar may produce false alarms from birds and similar small objects, sensor fusion enables enhanced robustness by combining information across modalities. We employ transformer-based fusion techniques to exploit cross-modal attention and scalability, enabling real-time UAV detection suitable for outdoor deployment in security, agricultural, and civilian applications. Our main contributions are:

- 1) A novel UAV detection and classification framework fusing radar and acoustic data via a transformer encoder for robust outdoor operation across security, agricultural, and civilian scenarios.
- 2) Experimental validation using frequency modulated continuous-wave (FMCW) radar and directional microphones with state-of-the-art performance. We publicly release our dataset¹ for reproducibility.
- 3) End-to-end processing of raw acoustic waveforms without domain transformations, reducing preprocessing overhead and minimizing information loss.
- 4) A joint training strategy with a custom gated loss function that simultaneously optimizes detection and classification, eliminating the need for two-stage pipelines.

II. RELATED WORKS

A. UAV Detection Modalities

A variety of sensing modalities have been explored for UAV detection, each with unique strengths and limitations. The

¹<https://huggingface.co/datasets/GevinduGanganath/DroneData>

choice of modality depends on detection range, environmental conditions, computational resources, and the target application.

Optical sensors: Vision-based systems employ cameras including RGB, thermal, and infrared to capture UAV imagery using handcrafted features (histogram of oriented gradients, scale-invariant feature transform) or deep learning classifiers such as convolutional neural networks (CNN) and YOLO variants [6]. Despite these advances, optical systems remain highly sensitive to fog, rain, smoke, and poor illumination, with effective range typically limited to a few hundred meters.

Acoustic sensors: Acoustic detection exploits the unique sound signatures generated by drone motors and propellers, which produce characteristic tonal and broadband frequency patterns [7]. Methods have evolved from simple frequency-domain analysis to deep learning classifiers. Kümmritz [8] investigated acoustic augmentation strategies including pitch shifting, harmonic distortion, and ambient noise integration to improve machine learning-driven classification robustness. Sun *et al.* [9] proposed a deep learning system using microphone arrays with beamforming, achieving detection up to 135 m. Al-Emadi *et al.* [10] explored generative adversarial network-based dataset enhancement for audio drone detection to address the scarcity of labeled acoustic data. Despite these advances, acoustic sensing remains fundamentally susceptible to wind, traffic, machinery, and other background noise sources.

Radio frequency (RF) sensors: RF detection analyzes wireless transmissions between drones and controllers to identify make, model, and flight mode [11], [12]. These methods leverage RF fingerprints, modulation patterns, and protocol characteristics. Frid *et al.* [13] fused RF and acoustic features using CNN and recurrent neural network (RNN) classifiers with improved low-signal-to-noise ratio (SNR) accuracy. Zheng *et al.* [14] demonstrated individual drone identification using raw I-Q data. However, RF detection faces fundamental limitations against encrypted, frequency-hopping, or autonomous drones without active communication links.

Radar sensors: Radar systems detect drones based on micro-Doppler signatures, radar cross section (RCS) characteristics [15]–[17] and vibration signatures of drones [18]. They offer all-weather operation, long detection range, and simultaneous range-velocity estimation. Morris and Hari [16] demonstrated UAV detection using millimeter-wave automotive radar. Sayed *et al.* [17] proposed classification via inverse synthetic aperture radar images with radar digital twins, and Wang *et al.* [19] applied deep learning to pulse-Doppler radar for UAV detection. However, radar may produce false positives from birds and similar small objects, and classification accuracy can be limited when using radar as a standalone modality.

B. Sensor Fusion Approaches

Due to the limitations of single-modality systems, multi-sensor fusion has become a prominent research direction. A comprehensive survey by Dong *et al.* [20] maps the evolution of anti-UAV research from single-modality models toward hybrid fusion architectures, revealing that modern counter-UAV systems increasingly combine radar, LiDAR, RGB, in-

frared, acoustic, and RF data to improve detection sensitivity and resilience. Multi-modal datasets such as MMAUD, which synchronize radar, LiDAR, audio, and vision data, reflect the consensus that no single sensor can reliably detect small, low-RCS drones in real operational environments.

Several notable fusion approaches exist. Svanström *et al.* [21] demonstrated benefits of combining visible, thermal, and acoustic sensors for real-time detection. Shen *et al.* [22] explored radar-acoustic fusion leveraging complementary range and frequency characteristics. Khan *et al.* [23] proposed SafeSpace MFNet for multi-feature drone detection. Fusion techniques are broadly categorized into: (i) feature-level fusion, where intermediate features from multiple sensors are integrated before classification; (ii) decision-level fusion, where independent classifier outputs are combined via ensemble or probabilistic methods; and (iii) deep learning-based fusion, which learns cross-modal representations end-to-end.

Transformer architectures have opened new possibilities for cross-modal fusion. Originally proposed for natural language processing [24], transformers have been adapted for autonomous driving [25], odometry estimation [26], multi-target tracking [27], and multi-modal event detection [28]. A survey on multi-sensor fusion [29] identifies transformers as the mainstream fusion framework across autonomous systems, drones, and agricultural engineering. Self-attention mechanisms effectively model long-range cross-modal dependencies, handle missing or corrupted sensor data, and scale to additional modalities with minimal architectural changes. Despite these successes, transformer-based fusion remains underexplored for UAV detection, particularly for radar-acoustic integration.

C. Summary

While individual sensing modalities show promise in controlled conditions, they face significant challenges under real-world scenarios with environmental variability, sensor noise, and adversarial countermeasures. Multi-modal sensor fusion has emerged as the most robust solution, and transformer-based approaches offer additional benefits in cross-modal learning and scalability. Our work builds on these insights by developing and validating a transformer-based radar-acoustic fusion model for UAV detection in outdoor environments, with applicability spanning security, agriculture, and civilian airspace management.

III. MEASUREMENT SETUP AND DATA COLLECTION

A. Hardware Configuration

We employ an FMCW radar (Texas Instruments AWR2243 Boost [30]) and a directional microphone for multi-modal data collection. The setup and processing chain are shown in Fig. 1. Table I presents the radar specifications. The radar operates in the 77 GHz band, providing high-resolution range and velocity measurements suitable for detecting small aerial targets. The directional microphone (sampling rate 16 kHz) operates in stereo mode, capturing acoustic signatures of drone motors and propellers.

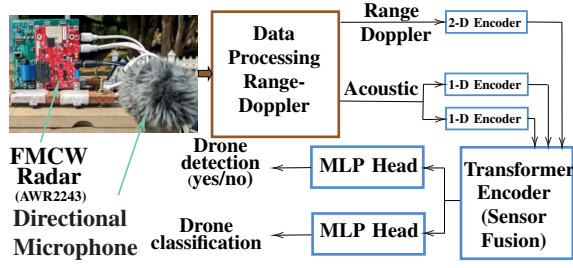


Fig. 1. Measurement setup with FMCW radar and directional microphone followed by the deep learning model. The radar captures range-Doppler signatures while the microphone records acoustic emissions.

B. Signal Processing

The FMCW radar transmits a continuously swept frequency signal and measures the beat frequency between transmitted and received signals to estimate range and velocity. We follow the standard pipeline: mixing the received signal with the transmitted signal to produce the intermediate frequency signal, then applying a 2D discrete Fourier transform using a fast Fourier transform algorithm across range and Doppler dimensions to construct the range-Doppler map [16], [31]. This map provides a two-dimensional representation of target range and radial velocity. A zero-Doppler filter suppresses ground clutter and stationary interference, and the continuous false alarm rate algorithm adaptively thresholds the map to extract target detections while maintaining controlled false alarm probability. For acoustic data, stereo recordings are converted to mono via channel averaging and normalized to compensate for amplitude variations due to flight distance and microphone characteristics. Crucially, no domain transformations (spectrograms, Mel-Frequency Cepstral Coefficients) are applied; the raw time-domain waveform is directly fed to the neural network, preserving all temporal and spectral information for end-to-end learning.

C. Data Collection Protocol

We collected data for five classes: no-drone, DJI Matrice 300 RTK, DJI Mavic 2 Enterprise Dual, DJI Phantom 4 Pro Plus, and DJI Phantom 4 Pro V2.0. These models represent a range of sizes, motor configurations, and acoustic profiles commonly encountered in practice. For each drone class, 2048 samples were collected; the non-drone class contains 2432 samples of ambient background.

Multiple flight scenarios were considered to ensure variability: (i) flights toward the radar at various altitudes and aspect angles (front, side, rear), (ii) cross-pattern flights at multiple heights, and (iii) random maneuvers including hovering, orbiting, and rapid directional changes. Radar binary files contain 256 frames per file (128 chirps $\times$ 256 samples each). Acoustic recordings are approximately 12 seconds long, captured synchronously with radar data. All data were collected outdoors to ensure realistic propagation, clutter, and noise conditions representative of security perimeters, agricultural fields, and open-area surveillance.

TABLE I
SPECIFICATIONS OF THE AWR2243 FMCW RADAR MODULE.

Parameter	Value	Parameter	Value
Start Frequency (GHz)	77	Samples per chirp	256
Chirps per frame	128	Sampling rate (MSPS)	10
Frame periodicity (ms)	40	Bandwidth (GHz)	0.76
Slope (MHz/ μ s)	29.98	Range resolution (m)	0.19
Velocity resolution (m/s)	0.15	No. of TX / RX antennas	1 / 1

IV. PROPOSED DEEP LEARNING MODEL

The proposed architecture, shown in Fig. 1, consists of three stages: modality-specific encoders, transformer-based fusion, and dual multi layer perceptron (MLP) heads for detection and classification.

A. Encoder Design

The encoders employ convolutional layers for hierarchical feature extraction, inspired by [32]. Residual connections are incorporated to mitigate vanishing gradients in deeper networks [33], enabling stable training. Each encoder integrates K squeeze-and-excitation blocks [34] that perform channel-wise recalibration by adaptively weighting feature channels based on global information content, emphasizing informative features while suppressing less relevant ones.

For acoustic data, we follow the multi-resolution approach of [35], using two parallel 1D encoders with different kernel sizes: a small kernel (size 7) captures fine-grained temporal details and high-frequency components, while a large kernel (size 107) captures broader temporal patterns and low-frequency characteristics. Outputs are combined via element-wise addition, creating a multi-resolution feature representation. An additional down-sampling convolution (kernel size 15) handles the large input dimensionality of raw audio. $K = 5$ for all 1D encoders.

For range-Doppler maps, a 2D encoder with $K = 4$ and progressively increasing kernel sizes of 3, 3, 5, and 7 captures multi-scale spatial features from fine-grained spectral patterns to broader structural characteristics. No down-sampling is required due to the compact input size. Kernel sizes were selected through systematic experiments to maximize detection and classification accuracy.

B. Transformer-Based Sensor Fusion

Rather than relying on simple concatenation or traditional fusion algorithms, we employ a transformer encoder [24] with eight attention heads, one layer, and embedding dimension 128 as the fusion module. Unlike RNN- or LSTM-based fusion, which imposes an artificial sequential ordering on the two modality embeddings, or static concatenation, which applies a fixed fusion rule regardless of signal quality, self-attention adaptively reweights each modality's contribution per sample, making it better suited to the dynamic, non-sequential radar-acoustic relationship in our setting. The transformer receives concatenated feature embeddings from both encoders. Through multi-head self-attention, the model learns cross-modal relationships, allowing radar features to attend to acoustic features

TABLE II
PERFORMANCE COMPARISON WITH LEE *ET AL.* [32].

Architecture	Detection		Classification	
	Acc. (%)	F1	Acc. (%)	F1
Lee <i>et al.</i> (DenseNet) [32]	75.00	0.7494	68.34	0.6733
Lee <i>et al.</i> (ResNet) [32]	75.00	0.7491	64.09	0.6179
Ours (Acoustic + Range-Doppler)	100.00	1.0000	100.00	1.0000

and vice versa. This soft feature selection is particularly advantageous when one modality is more informative than the other under varying environmental conditions.

Unlike two-stage architectures [32] requiring separate training of detection and classification models with independent encoders which demands significant effort and incurs extra computational cost at inference, we use two MLP heads operating on the shared fused representation: one for binary detection (drone vs. no-drone) and one for multi-class classification (identifying the specific drone type), enabling joint end-to-end training.

C. Joint Loss Function

To facilitate simultaneous training of both heads, the custom loss function is defined as:

$$\text{Loss} = \frac{1}{N} \sum_{n=1}^N X_{\text{ent}}(\hat{y}_{\text{det},i}, y_{\text{det},i}) + \lambda y_{\text{det},i} X_{\text{ent}}(\hat{y}_{\text{cls},i}, y_{\text{cls},i}), \quad (1)$$

where $X_{\text{ent}}(\hat{y}_{\text{det},i}, y_{\text{det},i})$ is the binary cross-entropy for detection with $y_{\text{det},i} \in \{0, 1\}$ indicating drone absence or presence. The classification term is gated by $y_{\text{det},i}$, ensuring it is active only when a drone is present ($y_{\text{det},i} = 1$). When no drone is detected, the classification loss is completely suppressed, preventing the model from learning spurious patterns from background samples. The parameter λ controls the relative importance of the classification loss, and the sum is normalized over mini-batch size N .

D. Class Imbalance and Training Configuration

Joint training introduces a class imbalance: merging four drone classes for the detection head creates a 4:1 imbalance. We up-sample the non-drone class to balance detection training. This does not affect classification balance because the gating mechanism zeros out all non-drone contributions:

$$X_{\text{ent}}(\hat{y}_{\text{cls}}, y_{\text{cls}}) = \begin{cases} X_{\text{ent}}(\hat{y}_{\text{cls}}, y_{\text{cls}}) & y_{\text{det}} = 1 \\ 0 & y_{\text{det}} = 0. \end{cases} \quad (2)$$

The effective class distribution seen by the classification head remains balanced across the four drone types, allowing simultaneous balanced training for both heads without separate sampling strategies.

The final model has 15 M parameters and 68.5 GFLOPS. Training uses an NVIDIA RTX 3090 for 60 epochs, batch size 64, Adam optimizer with learning rate 5×10^{-5} , weight decay 0.4, and dropout 0.4. Early stopping based on validation loss prevents over-fitting. Following the procedure in [36] we employed trial-wise non-overlapping splitting of 70%/15%/15%

TABLE III
ACOUSTIC-ONLY MODEL COMPARISON ON DATASET FROM [7].

Architecture	Input	Params (M)	End-to-end	Acc. (%)	F1
ResNet101 [33]	Mel spec.	45	No	94.68	0.956
AST [37]	Mel spec.	87	No	85.11	0.848
Ours	Raw audio	23	Yes	98.87	0.989

for training, validation, and testing with stratified sampling to preserve class proportions.

V. EXPERIMENTAL RESULTS

We present comprehensive experimental results to validate the effectiveness of the proposed approach. The evaluation covers comparisons with existing methods, single-modality ablation, and noise robustness analysis.

A. Comparison with Existing Methods

Table II compares our model with Lee *et al.* [32] on our dataset using DenseNet and ResNet backbones. Our method achieves perfect accuracy (100%) and F1-score (1.0) for both detection and classification, significantly outperforming [32] which reaches only 75% detection and 64–68% classification accuracy. This substantial improvement stems from three factors: (i) the multi-resolution convolutional encoders capturing complementary features from both modalities, (ii) the transformer-based fusion learning optimal cross-modal interactions, and (iii) the joint training strategy enabling both heads to benefit from shared representations.

B. Acoustic-Only Model Evaluation

To assess the standalone capability of our encoder architecture, we evaluate the acoustic-only model on the DronePrint dataset [7]. As shown in Table III, our model achieves 98.87% accuracy and 0.989 F1-score using raw audio with only 23M parameters surpassing ResNet101 (94.68%, 45M params) and audio spectrogram transformer (AST) [37] (85.11%, 87M params) by significant margins. Crucially, our model operates end-to-end without spectrogram conversion and uses substantially fewer parameters, demonstrating its potential for resource-constrained edge deployment in security and agricultural monitoring.

C. Effect of Environmental Noise

Despite strong acoustic-only results, acoustic sensing is highly susceptible to environmental noise a critical limitation for practical outdoor deployment where wind, machinery, and traffic are prevalent. To quantify this, we conduct a noise robustness study by adding Gaussian noise at controlled SNR levels to our captured acoustic dataset.

Table IV presents the results. For the acoustic-only model, classification accuracy drops catastrophically from 84.33% (no noise) to 25.27% at 6 dB SNR near random chance for a four-class problem. Detection also degrades from 99.88% to 77.76%. These results clearly demonstrate that standalone acoustic sensing is insufficient for robust operation in noisy

TABLE IV
EFFECT OF NOISE ON DIFFERENT MODALITY CONFIGURATIONS.

SNR (dB)	Acoustic Only		Acoustic + Range-Doppler	
	Det. Acc. (%)	Cls. Acc. (%)	Det. Acc. (%)	Cls. Acc. (%)
6	77.76	25.27	93.56	39.13
12	87.36	27.70	95.87	49.94
18	91.25	31.96	97.21	58.44
24	96.72	39.49	99.87	59.66
No added noise	99.88	84.33	100.00	100.00

environments typical of security patrols, agricultural fields, or urban surveillance.

The fused system shows substantially improved robustness. At 6 dB, detection accuracy improves by 15.8 percentage points (77.76% $\rightarrow$ 93.56%) and classification by 13.86 points (25.27% $\rightarrow$ 39.13%). At 24 dB, the fused system reaches near-perfect detection (99.87%). These gains are attributable to radar features being unaffected by acoustic noise, providing stable range-Doppler cues that anchor the fused representation even when acoustic features are severely degraded. The results validate the multi-modal approach for real-world deployment in environments with variable and unpredictable noise conditions.

D. Modality Ablation Study

Table V compares different modality combinations. For detection, the fused system achieves perfect accuracy (100%). Acoustic-only performs near-perfectly (99.88%), indicating strong discriminative power for binary presence detection. Range-Doppler alone drops to 76.55%, indicating radar features are less effective in isolation at the tested ranges.

For classification, the gap widens further. The fused system achieves 100%, acoustic-only drops to 84.33%, and range-Doppler alone yields only 16.28% suggesting single-antenna micro-Doppler signatures are insufficiently distinctive to differentiate drone models in isolation. However, when combined with acoustic data, radar features provide complementary cues that eliminate residual classification errors, achieving perfect performance. These findings highlight the complementary nature of the two modalities: acoustic features provide primary discriminative power for both tasks, while radar features contribute robustness under noisy conditions and supply additional information that perfects the fused classification.

E. Discussion

The experimental results yield several key insights for designing practical UAV detection systems. First, multi-modal fusion consistently outperforms any single modality across all evaluation metrics and noise conditions, validating the fundamental hypothesis of this work. Second, the transformer-based fusion mechanism effectively learns to exploit cross-modal complementarity, achieving perfect scores even though neither modality achieves perfect classification independently. Third, end-to-end processing of raw acoustic waveforms proves highly effective, confirming that our convolutional encoders autonomously learn relevant features without manual feature engineering or domain transformations. Fourth, the joint

TABLE V
PERFORMANCE WITH DIFFERENT MODALITY COMBINATIONS.

Modalities	Detection		Classification	
	Acc. (%)	F1	Acc. (%)	F1
Acoustic + Range-Doppler	100.00	1.0000	100.00	1.0000
Acoustic only	99.88	0.9988	84.33	0.8597
Range-Doppler only	76.55	0.7654	16.28	0.0885

detection-classification paradigm with the gated loss function avoids the complexity and error propagation of two-stage pipelines, making the system practical for real-time deployment in resource-constrained scenarios common in agricultural monitoring and field security applications.

It is also worth noting that classification accuracy under severe noise remains low even with fusion: 39.13% at 6 dB SNR is only 14 percentage points above the 25% random-chance baseline for a four-class problem. We therefore qualify our robustness claim: fusion substantially improves classification relative to the acoustic-only model (25.27% $\rightarrow$ 39.13%, a 13.86-point gain) without closing the gap to clean-condition performance, and this improvement should not be interpreted as the fused classifier being reliable under extreme noise. Detection, by contrast, remains genuinely robust across all tested SNR levels (93.56% at 6 dB, 99.87% at 24 dB), and we consider this the operationally critical capability for deployment, since correctly flagging the presence of an unauthorized drone is the necessary first step regardless of whether its specific type can be identified under extreme noise.

VI. CONCLUSION AND FUTURE WORK

We investigated drone detection and classification in outdoor environments using a multi-modal approach combining FMCW radar and acoustic sensing with transformer encoder-based fusion. The framework demonstrates broad applicability across security surveillance, agricultural airspace protection, infrastructure monitoring, and civilian drone management. Our approach significantly outperforms existing methods while utilizing fewer model parameters, and the noise analysis confirms that radar-acoustic fusion substantially improves robustness over acoustic-only systems, with detection accuracy improvements of up to 15.8 percentage points at low SNR.

Several promising directions remain for future work. First, we plan to validate the approach for long-range detection using high-power radar paired with sensitive microphone arrays, extending the operational envelope for large-area security and agricultural monitoring. Second, evaluation under real-world environmental noise including wind, rain, and urban backgrounds beyond synthetic Gaussian noise will strengthen practical applicability. Third, incorporating additional modalities such as thermal imaging, RF sensing, or LiDAR could further enhance robustness and classification granularity. Fourth, investigating lightweight architectures and quantization for edge deployment on embedded platforms would facilitate operation on autonomous security drones and agricultural monitoring stations. Finally, expanding the dataset to include a broader

range of drone types, flight altitudes, and environmental conditions will improve generalizability. A related direction is extending the classification head beyond its current closed-set formulation to support open-set recognition, allowing the system to flag a previously unseen drone type as “unknown” rather than forcing it into one of the trained classes. This could be achieved through confidence-based rejection.

ACKNOWLEDGMENT

We extend our sincere gratitude to the Sri Lanka Air Force for granting access to their drones and facilities. The computational resources used in this work were funded by the Accelerating Higher Education Expansion and Development (AHEAD) Operation; grant No. 6026-LK/8743-LK of the Ministry of Higher Education of Sri Lanka funded by the World Bank.

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